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Relationship between university students' personalities and e-learning engagement mediated by achievement emotions and adaptability

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Abstract

Since the outbreak of the pandemic, a large number of university students have been required to adapt to e-learning. E-learning engagement, a key indicator of academic success, is thus a great concern of educational partitioners and researchers. Although previous studies have identified various determinants of e-learning engagement, there is a paucity of research that examines the associations between university students' personalities and their engagement in e-learning. Moreover, the mechanisms underlying the personality-engagement relationship are still poorly understood. The present study used the five-factor model of personality as the main theoretical framework to explore how students with different personality traits engaged in e-learning. Additionally, it examined whether achievement emotions and adaptability mediated the personality-engagement relationship in the e-learning context. A sample of 1004 students enrolled at Guizhou University participated in an online survey to collect data for the study. Employing structural equation modeling, the findings unveiled several significant results: (1) extroversion, agreeableness, openness to new experiences, and conscientiousness exhibited positive associations with e-learning engagement, (2) neuroticism demonstrated a negative relationship with e-learning engagement, (3) the mediating effect of enjoyment as an achievement emotion was observed between personality traits (excluding neuroticism) and e-learning engagement, (4) adaptability played a mediating role in the relationship between personality traits (excluding conscientiousness and neuroticism) and e-learning engagement, and (5) the negative achievement emotion of anxiety did not operate as a mediator between personality traits and e-learning engagement. This study enriches the understanding of the relationship between personality and engagement in the emerging field of e-learning. Moreover, it offers a fresh perspective on how to investigate mechanisms underlying the relationship between personality and engagement in the e-learning

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context. The findings could provide a basis for instructors who wish to deploy emotional and adaptability interventions to increase university students' engagement in e-learning.

Keywords University students · e-learning engagement · Personality · Achievement emotion · Enjoyment · Anxiety · Adaptability

1 Introduction

The unpredictable pandemic has led to a considerable shift in learning mode, where e-learning is becoming increasingly prevalent around the world (Yu, 2022; Zhang et al., 2023). Although the pandemic has subsided, the mass adoption of e-learning has exerted a profound effect on the systems of higher education (Yan et al., 2021). University students around the world are still committed to e-learning even though the pandemic has abated. Acknowledging the significance of e-learning in higher education, scholars and educational practitioners have devoted considerable attention to analyzing and supporting the success of university students in e-learning settings (Adeshola & Agoyi, 2022; Alon et al., 2023). Emerging research has posited that students' engagement in e-learning plays a crucial role in their ultimate success (Chen et al., 2021). Specifically, their level of e-learning engagement has been found to exert a substantial impact on various outcomes, including learning effectiveness (Poon et al., 2022), academic performance (Yu, 2021), and overall achievement (Slof et al., 2021).

Nonetheless, it is worth noting that e-learning may not be suitable for every university student (Baruth & Cohen, 2023). Several studies have found that many students encountered difficulties while adapting to the e-learning environment, leading to limited levels of engagement (Ewing & Cooper, 2021; Wester et al., 2021; Wu & Teets, 2021; Zhao et al., 2022). Unfortunately, this has resulted in a low completion rate of online courses (Martin & Borup, 2022), compromised learning effectiveness (Yu et al., 2022), and reduced satisfaction with the learning experience (Liu et al., 2022). Consequently, there is an urgent need to comprehend the various factors that influence students' engagement in e-learning to enhance their success in e-learning (Sadoughi & Hejazi, 2021; Wang et al., 2022).

Personality traits (i.e., extraversion, agreeableness, conscientiousness, neuroticism, and openness to a new experience; McCrae & Costa, 1987) were found to be closely related to students' learning engagement in the e-learning context (Quigley et al., 2022). Students could learn at their own pace in e-learning (Varela et al., 2012). However, not all students had a high level of engagement in the e-learning environment. Some students were unfocused and disengaged in e-learning, while others still behaved in an organized way and were immersive in e-learning (Li et al., 2021). Students had different learning behaviors and reactions to e-learning, depending on their personality traits (Lee & Wu, 2022). Nevertheless, previous studies have paid inadequate attention to the relationship between university students' personalities and e-learning engagement (Moreira et al., 2021). Examining the nexus between personality traits and engagement in online learning can enable educational practitioners to effectively identify university

students requiring assistance in effectively participating in e-learning (Quigley et al., 2022). The first research question in this study is thus "What is the relationship between university students' personality and e-learning engagement?"

The relationship between personality and e-learning engagement might be more complex (Bosselut et al., 2020). Personality was a distal factor that indirectly influenced learning engagement through other factors (Woods & Sofat, 2013). An empirical study conducted by Chen et al. (2021) supported the argument and showed that self-efficacy and student–teacher relationships fully mediated the personality-engagement relationship in the e-learning context. According to the control-value theory and cognitive-developmental theory, achievement emotions and adaptability could be potential mediators during the learning process (Martin et al., 2013; Pekrun, 2006). Achievement emotions could be defined as emotions in relation to achievement activities and outcomes (Pekrun, 2006). Adaptability refers to an individual's ability to constructively adjust and regulate cognition, emotions, and behaviors in response to new and changing environments (Fredrickson, 2001).

Nevertheless, there is a notable paucity of empirical research focusing on the mediating role of achievement emotions and adaptability in the relationship between personality and e-learning engagement. Research on the mediating roles of achievement emotions and adaptability could provide a basis for the adoption of emotional interventions and adaptability interventions for online learners. Therefore, the second research question in this study is "Could achievement emotions mediate the relationships between student personality and e-learning engagement?", and the third research question in this study is "Could adaptability mediate the relationship between student personality and e-learning engagement?

This current study offers significant contributions to the field. Firstly, it investigates the variations in e-learning engagement among university students with diverse personality traits, shedding light on specific personality types that may require additional support in effectively participating in e-learning (Quigley et al., 2022). This study provides supplementary evidence on the relationship between personality and engagement in the e-learning setting, distinct from the findings pertaining to traditional learning contexts. Secondly, this study introduces and verifies the intermediary role of achievement emotions and adaptability in e-learning engagement, thereby highlighting the newfound significance of these factors within e-learning environments. By exploring achievement emotions and adaptability, this study widens the scope of focus within the e-learning context. Thirdly, this study offers valuable insights into the customization of online courses and the provision of personalized e-learning experiences based on specific personality types. By tailoring e-learning approaches to individual personality traits, this study contributes towards enhancing university students' learning engagement and fostering sustainable e-learning practices.

2 Literature review

2.1 Learning engagement

Learning engagement in this study was operationally defined as the physical and mental effort that students devoted to e-learning. Learning engagement is a multidimensional construct, which consists of behavioral engagement, cognitive engagement, and emotional engagement (Fredricks et al., 2004). Behavioral engagement refers to students' active participation in e-learning. Emotional engagement refers to students' feelings and emotions concerning their instructor, classmates, and activities in e-learning. Cognitive engagement refers to students' efforts to keep up with e-learning (Meyer & Turner, 2002).

There has been a growing body of evidence to suggest that learning engagement could be attributed to individual factors. Students with higher self-efficacy showed higher engagement than those with less (Jung & Lee, 2018). Having a lower level of stress, undergraduates were more likely to engage in learning activities. Student belief was an important driving factor in learning engagement (Zheng et al., 2023). Earl et al. (2023) surveyed 361 adolescents and found that psychological need satisfaction was positively associated with cognitive engagement. It was noteworthy that personality traits were found to be closely related to e-learning engagement (e.g., Audet et al., 2021; Chen et al., 2021; Quigley et al., 2022).

2.2 Personality

Personality traits are endogenous basic tendencies to evaluate the similarities and differences between people (Lee & Wu, 2022). Personality in this study could be defined as individual differences in behaviors, cognition, emotions, and habits, which could remain stable over time. One of the most well-accepted models of personality is the five-factor model (McCrae & Costa, 1987). According to the five-factor model (FFM) of personality, personality could be classified in terms of five dimensions, including agreeableness, conscientiousness, extroversion, neuroticism, and openness to a new experience (Goldberg, 1993; John & Srivastava, 1999; McCrae & Costa, 1987).

Previous findings about the personality-engagement relationship have been inconsistent and contradictory. Some studies found that students' personality traits were strongly related to their learning engagement (Wildermuth et al., 2013). Conscientiousness, agreeableness, extroversion, and openness to a new experience were generally found to be positively associated with student engagement in offline learning (Komarraju & Karau, 2005; Qureshi et al., 2016). By contrast, neuroticism appeared to be negatively related to school engagement (Sulea et al., 2015). However, in a study conducted by Closson and Boutilier (2017), it showed that no significant correlations were found between neuroticism and academic engagement.

Despite the popularity of online education, far too little attention has been paid to the relationship between personality and learning engagement as well as the mediating effects of achievement emotions and adaptability on the personality-engagement

relationship in the e-learning context. In response to research gaps, this study aimed to use the five-factor model of personality as the theoretical framework to explore the personality-engagement relationship in the e-learning context. Moreover, integrating the control-value theory and cognitive-developmental theory with the five-factor model of personality, this study set out to analyze whether achievement emotions and adaptability could act as mediators between personality and e-learning engagement. Therefore, this study contributes to the existing knowledge by using the five-factor model of personality to explain how and why university students with different personality traits differed in e-learning engagement.

3 Hypotheses development research model

3.1 The role of agreeableness

Agreeable individuals are characterized as kind, cooperative, helpful, compliant, and sympathetic (McCrae & John, 1992). Individuals with stronger agreeableness tended to foster teamwork and collaboration, maintain trusting relationships, and create positive environments (Wilmot & Ones, 2022). Due to their friendly nature, students with strong agreeableness have usually behaved in a desirable way in e-learning (Bhagat et al., 2019). They tended to build good relationships and receive emotional support from teachers and peers in the e-learning environment (Chen et al., 2022). Therefore, they were inclined to actively interact with peers and teachers in e-learning processes (Yu, 2021). The online interaction, based on social constructivism learning theory, could facilitate students' cooperative learning and learning engagement in e-learning (Nungu et al., 2023). We thus proposed the following hypothesis:

H1: Agreeableness is positively associated with e-learning engagement.

3.2 The role of conscientiousness

Highly conscientious individuals are labeled as responsible, diligent, persistent, organized, and goal-oriented (McCrae & John, 1992). Highly conscientious students were more likely to set learning goals and then devoted their efforts to achieving learning goals (Spielmann et al., 2022). Despite a lack of teacher supervision, highly conscientious students did not indulge in online games and entertainment because they had greater resistance to external disturbances in e-learning. They still could remain fully engaged and perform well in e-learning due to their self-discipline, persistence, and diligence (Closson & Boutilier, 2017; Sak, 2021). We thus hypothesized that:

H2: Conscientiousness is positively associated with e-learning engagement.

3.3 The role of extroversion

Extroverted individuals are characterized as sociable, outgoing, active, and gregarious (McCrae & John, 1992). Extroverted students, who put a high value on social relationships, were willing to participate in various e-learning activities (Mihai et al., 2022). By contrast, introverted students may have an uneasy feeling when expressing their ideas in e-learning activities (Iterbeke & De Witte, 2021). Therefore, extroverted students were more likely to engage in various learning activities than introverted ones (Komarraju & Karau, 2005; Wang et al., 2013). However, an empirical study by Sulea et al. (2015) reached different conclusions, finding that extroverted students did not show higher levels of learning engagement than introverted ones. Another supportive research was that there was no significant difference in the levels of learning engagement between extroverts and introverts (Quigley et al., 2022). Therefore, we proposed the following hypothesis:

H3: Extroversion is positively associated with e-learning engagement.

3.4 The role of neuroticism

Individuals with strong neuroticism are highly emotional, depressed, and tense (McCrae & John, 1992). Students high in neuroticism tended to become more stressed when they were uncertain about lectures' explanations and assignments in courses (Yusoff et al., 2021). They experienced more burnout and technostress after spending many hours on online courses. Worse, they might have difficulties in coping with negative emotions appropriately (Izhar et al., 2022), It could lead neurotic students to lack attention, energy, and interest in e-learning (de la Fuente et al., 2020). We thus proposed the following hypothesis:

H4: Neuroticism is negatively associated with e-learning engagement.

3.5 The role of openness to a new experience

E-learning could provide university students with more new educational resources as well as flexibility in time and location (Archambault et al., 2022). The benefits of e-learning could intrinsically motivate students, especially those seeking new experiences or chances to develop knowledge and skills. Students with openness to a new experience, who are intellectually curious and behaviorally flexible, were more receptive to online education and were intrinsically motivated to participate in e-learning (Komarraju & Karau, 2005). They had a more positive perception of online education (Keller & Karau, 2013). We thus proposed the following hypothesis:

H5: Openness to a new experience is positively associated with e-learning engagement

3.6 The mediating roles of achievement emotions

According to the control-value theory of achievement emotion, Achievement emotions are complex and multifaceted phenomena involving action tendencies, cognitive processes, and subjective feelings (Pekrun, 2017). Achievement emotions differ from other emotions in that they may involve object-focused, appraisal-driven, and multicomponent psychological processes (Peng et al., 2020). Achievement emotions could be generally categorized into positive emotions and negative emotions (Pekrun, 2006). Positive emotions could be further classified into activating and deactivating emotions, such as enjoyment and relaxation. Negative emotions could be further grouped into activating and deactivating negative emotions, such as anxiety and boredom. It was necessary to further investigate specific achievement emotions, given that each may have a different effect on students' learning outcomes (Wang et al., 2017). This study focused mainly on enjoyment and anxiety because they were experienced most frequently by students (Liu et al., 2021a; Pekrun et al., 2002).

A growing body of literature has shown a strong and consistent association between personality and achievement emotions, especially enjoyment and anxiety. Conscientiousness has been consistently associated with enjoyment (Sander & de la Fuente, 2022). Extroverted students were more active, enthusiastic, and cheerful. Extroverted students tended to experience more enjoyment in the learning process than introverted ones (Sulea et al., 2015). Moreover, undergraduate nursing students with similar personality traits, such as agreeableness and openness to a new experience, had a lower level of anxiety (Santo et al., 2022). Conversely, highly neurotic undergraduate students tended to use self-critical coping strategies associated with feelings of anxiety (de la Fuente et al., 2020).

Besides, enjoyment and anxiety could exert a significant influence on learning engagement (Fredricks et al., 2004). Based on the control-value theory of achievement emotions, it has been posited that experiencing enjoyment can positively impact various aspects of student engagement in learning, including focus, productivity, interest, cognitive resources, and persistence (Dewaele & Li, 2021; Fredrickson, 2001). Conversely, anxiety, as postulated by the same theoretical framework, is believed to heighten students' cognitive load and deplete their working memory resources, resulting in decreased levels of learning engagement (Deng et al., 2022; Pekrun et al., 2011). Taken all together, we proposed the following hypotheses:

H6: In the e-learning context, enjoyment could mediate the relationship between agreeableness and engagement (H6a), conscientiousness and engagement (H6b), extroversion and engagement (H6c), neuroticism and engagement (H6d), and openness to a new experience and engagement (H6e).

H7: In the e-learning context, anxiety could mediate the relationship between agreeableness and engagement (H7a), conscientiousness and engagement (H7b), extroversion and engagement (H7c), neuroticism and engagement (H7d), and openness to a new experience and engagement (H7e).

3.7 The mediating role of adaptability

The existing literature has revealed that personality could be a determining factor of adaptability. Students with different personalities might have different adaptability to learning environments (Wang et al., 2023). Highly conscientious students could better adapt to learning environments, whereas highly neurotic ones had great difficulties in adapting to learning environments (Martin et al., 2013). In e-learning environments, students scoring higher on agreeableness, openness to a new experience, and extroversion had a higher level of adaptability (Besser et al., 2022).

According to Piaget's (1952) cognitive-developmental theory, individuals possess inherent abilities to adjust and adapt to novel and evolving learning environments, thus facilitating their psychological development (Wang et al., 2023). Numerous empirical studies provide support for this notion and have identified adaptability as a significant determinant of learning engagement (Collie et al., 2017). Secondary school students with high adaptability tended to solve mathematical problems, engaging more fully in mathematics (Collie & Martin, 2017). Martin et al. (2017) also found that high school student adaptability was strongly related to motivation and engagement across China, North America, and the UK. Similarly, university students with high adaptability had high behavioral engagement and completion rates (Holliman et al., 2018). In addition, a study conducted by Audet et al. (2021) showed that students with openness to a new experience flexibly adapted to e-learning, thereby having a high level of engagement in online classes. Therefore, we expected that adaptability could mediate the personality-engagement relationship in the e-learning context. Therefore, we proposed the following hypotheses:

H8: In the e-learning context, adaptability mediates the relationship between agreeableness and engagement (H8a), conscientiousness and engagement (H8b), extroversion and engagement (H8c), neuroticism and engagement (H8d), and openness to a new experience and engagement (H8e).

Taken together, this study not only explored the structural relationship between personality traits and e-learning engagement but also examined the mediating roles of enjoyment, anxiety, and adaptability in the personality-engagement relationship. Figure 1 shows the proposed model.

4 Methods

4.1 The quantitative phrases

4.1.1 Participants

This research was approved by Research Ethics Board and consistent with APA ethical standards. During the timeframe encompassing November and December 2022, the present study employed WeChat as a means of communication with the participants. Participants voluntarily signed the informed consent form prior

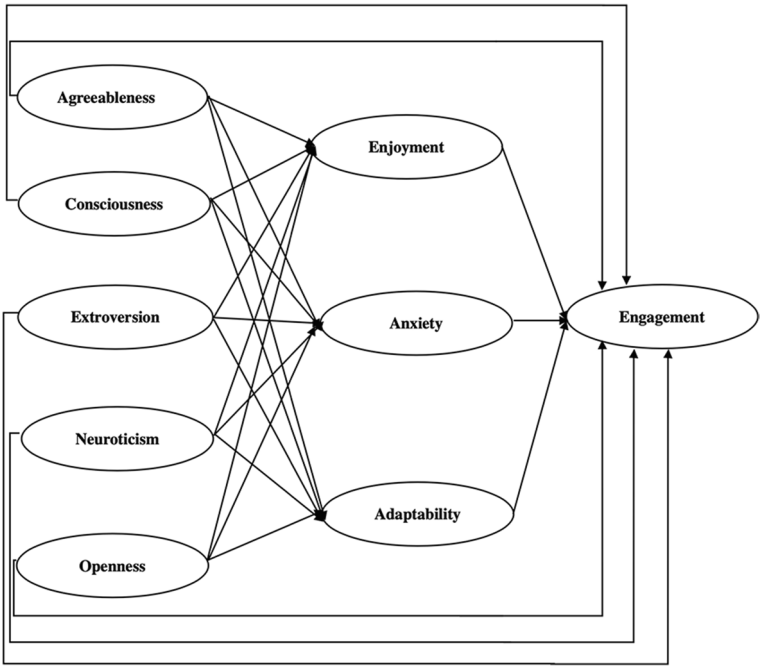


Fig. 1 The proposed model

Table 1 Details of participants' demographic information

		Frequency	Percent (%)
Gender	Male	228	22.7
	Female	776	77.3
Age	17 years old and younger	36	3.59
	18–22 years old	627	62.45
	23–27 years old	325	32.37
	28 years old and older	16	1.59
The educational level	Undergraduates	655	65.2
	Postgraduates	349	34.8
Total		10,041	100

to their participation. They were informed that they could withdraw from the research at any moment and the data would remain confidential. Each participant received rewards for their voluntary participation. This study randomly recruited 1004 students from Guizhou University in China. Females (776, 77.3%) outnumbered males (228, 22.7%). Of these participants, 655 were undergraduates and 349 were postgraduates, ranging from 17 to 28 years old. Table 1 displays the details of all participants' demographic information.

4.1.2 Measurements

The scale adapted from Reeve and Tseng (2011) were employed for measuring

Table 2 Summary of the factor loading, AVE, CR, and Cronbach's alpha results

Constructs	Factor loading	CR	AVE	Cronbach's alpha
EX				
EX1	0.709	0.842	0.571	0.794
EX2	0.774			
EX3	0.803			
EX4	0.734			
NEURO				
NEURO1	0.698	0.803	0.504	0.798
NEURO2	0.730			
NEURO3	0.712			
NEURO4	0.700			
CON				
CON1	0.638	0.799	0.501	0.742
CON2	0.788			
CON3	0.758			
CON4	0.635			
AGR				
AGR1	0.741	0.784	0.476	0.798
AGR2	0.648			
AGR3	0.674			
AGR4	0.693			
OPEN				
OPEN1	0.806	0.862	0.601	0.828
OPEN2	0.813			
OPEN3	0.796			
OPEN4	0.703			
ANX				
ANX1	0.805			
ANX2	0.801		0.568	
ANX3	0.779			
ANX4	0.783	0.887		0.860
ANX5	0.683			
ANX6	0.658			
EN				
EN1	0.754	0.877	0.544	0.803
EN2	0.778			
EN3	0.706			
EN4	0.693			
EN5	0.654			
EN6	0.825			

Table 2 (continued)

Constructs	Factor loading	CR	AVE	Cronbach's alpha
AD				
AD1	0.530	0.777	0.470	0.705
AD2	0.747			
AD3	0.749			
AD4	0.694			
E				
E1	0.787			
E2	0.636			
E3	0.660	0.848	0.530	0.838
E4	0.718			
E5	0.822			
ALL				**0.933**

NEURO neuroticism; *EX* extroversion; *OPEN* openness to a new experience; *AGR* agreeableness; *CON* consciousness; *E* engagement; *ANX* anxiety; *AD* adaptability; *EN* enjoyment. Bold values represent square root values of AVE. *CR* composite reliability; *AVE* average variance extracted

e-learning engagement. It consists of five items, for example, "I listen carefully in online classes," and "Before I begin to study online, I think about what I want to get done." All items are scored on a 1 to 5 Likert scale, ranging from strongly disagree to strongly agree. Previous studies have demonstrated the reliability and validity of the scale (e.g., Sadoughi & Hejazi, 2021). As can be seen from Table 2, the scale in this study was considered reliable ($\alpha = 0.838$).

This study focused on achievement emotions of enjoyment and anxiety. Enjoyment and anxiety constructs were measured using the subscales of the Achievement Emotions Questionnaire-Short (AEQ-S) (Collie et al., 2017). The subscales have been proven to be internally reliable and factorial valid in various educational contexts (e.g., Yilmaz et al., 2022). There are a total of six items to determine enjoyment, for example, "I enjoy being in online courses." There are six items to determine anxiety, for example, "I get tense in e-learning." All items are scored based on a five-point Likert scale. As shown in Table 2, Cronbach's alpha values in this study reached a good level on the scales of enjoyment ($\alpha = 0.803$) and anxiety ($\alpha = 0.860$).

To measure adaptability, valid scales adapted from Martin et al. (2012) were utilized. Sample items are "I am able to revise the way I think about a new situation to help me through it," and "I am able to think through a number of possible options to assist me in e-learning." Some previous studies provided convincing evidence of the reliability and validity of this scale (e.g., Collie et al., 2018). Four items are scored based on a 5-Likert scale. The Cronbach's alpha value in this study was 0.705, indicating an adequate level of reliability (see Table 2).

Personality measures were obtained from the Big Five Inventory developed by John and Srivastava (1999). Responses to 20 items could be used to obtain the level of extroversion, neuroticism, conscientiousness, agreeableness, and openness to a

new experience. Sample item includes "I see myself as someone who has a forgiving nature." For all items, a five-point Likert scale was used. Numerous studies have used BFI due to its reliability (e..g, Yu, 2021). Cronbach's alpha values for extroversion, neuroticism, conscientiousness, agreeableness, and openness to a new experience are 0.794, 0.798, 0.742, 0.798, and 0.828 respectively, indicating sufficient reliability (see Table 2).

4.1.3 Data analyses

Data analyses were performed using SPSS (version 20) program and Amos (version 26) software. Using SPSS, we calculated the descriptive statistics and Cronbach's alpha value. Following Anderson and Gerbing's (1988) suggestions, we adopted a two-step approach to testing the proposed model. To validate the measurement model, we initially conducted confirmatory factor analysis (CFA), with a maximum likelihood parameter estimation method. Then, structural equation modeling (SEM) was performed to assess the multiple-mediation model. SEM emerges as a more versatile approach for estimating the associations between variables when compared to multiple regression techniques (Wu et al., 2018). In addition, SEM offers the advantage of accounting for measurement error and producing more precise estimates of structure coefficients (Zyphur et al., 2023). Moreover, empirical investigations have consistently demonstrated the effectiveness of SEM in exploring the nexus between personality and engagement (Chen et al., 2021; Sulea et al., 2015). The statistical significance of each effect was estimated using the bias-corrected bootstrapping procedure with 95% confidence intervals (BC 95% CI) and 5000 bootstrapping samples (Preacher & Hayes, 2008).

Several fit indexes could be used to estimate the model fit (see Table 4), including the Chi-square Test (χ^2), Ratio of Chi-square to the Degrees of Freedom (CMIN/DF), Root Mean Square Error of Approximation (RMSEA), Comparative-Fit Index (CFI), Tucker-Lewis Index (TLI), Standardized Root Mean Square Residual (SRMR), Goodness-of-Fit Index (GFI), and Adjusted Goodness-of-Fit index (AGFI). A model could be considered to be an adequate fit to the quantitative data at a value of ≤ 0.05 for RMSEA, a value of ≥ 0.9 for CFI, a value of ≥ 0.9 for TLI, a value of ≤ 0.05 for SRMR, a value of ≥ 0.9 for GFI, and a value of ≥ 0.9 for AGFI (Bentler & Bonett, 1980; Hu & Bentler, 1999).

4.2 The qualitative phrase

4.2.1 Semi-structured interviews

Semi-structured interviews were employed to delve into the qualitative aspects underlying the quantitative findings, aiming to provide a comprehensive understanding of the phenomenon under investigation (Creswell & Clark, 2017). The interview protocol consisted of two distinct sections. In Section One, interviewees were requested to complete consent forms and provide basic demographic information.

Section Two encompassed a series of pertinent inquiries pertaining to personality traits, e-learning engagement, anxiety, adaptability, and enjoyment. Sample questions included, but were not limited to, "What types of personality traits do you perceive yourself as having?", "To what extent do you feel engaged in e-learning?", "How adaptable do you consider yourself to be in the context of e-learning?", "To what degree do you find e-learning to be enjoyable?", and "Do you experience feelings of anxiety when engaging in e-learning?"

A purposive sampling technique was employed to recruit university students who had engaged in e-learning during the pandemic. Prior to the face-to-face interviews conducted in July 2023, all participants voluntarily provided their informed consent by signing consent forms. In order to address language preferences, interviewees were offered the opportunity to express their perceptions and opinions in either English or Chinese. The interview duration varied, ranging from 22 to 68 min. Saturation of data was achieved after conducting 18 interviews, as no novel themes or topics emerged in subsequent interviews (Saunders et al., 2018). The entire interview process was audio-recorded, and the interview data were transcribed verbatim to facilitate further analysis.

4.2.2 Data analysis

The interview transcripts underwent a thematic analysis method as recommended by Braun and Clarke (2006). Initially, the researchers familiarized themselves with the data by reviewing the transcribed texts multiple times, a process that enabled them to ensure accuracy and eliminate irrelevant content (Braun & Clarke, 2006). Subsequently, the researchers coded the transcribed data, identifying and formulating initial codes. In the next step, the researchers sought to amalgamate the codes into broader themes, capturing those that were pertinent to the research objectives. Following this, the researchers reviewed, verified, and deliberated on the themes. Moreover, they clarified and defined the names of the themes. Ultimately, the researchers presented a summary of the qualitative data and generated a concise academic report (Braun & Clarke, 2006).

5 Results

5.1 Results from the questionnaire

5.1.1 The measurement model

The present study assessed the measurement model based on three criteria, including reliability, convergent validity, and discriminant validity (Hair et al., 2014). The reliability could be assessed using composite reliability (CR) and Cronbach's alpha value. CR values should exceed 0.7 (Fornell & Larcker, 1981), while Cronbach's alpha values should exceed 0.7 (Hair et al., 2014). The convergent validity could be evaluated using factor loading and average variance extracted (AVE). Ideally, the factor loading and AVE should exceed 0.5 (Hair et al., 2014). Fornell Larker

Criterion is a widely accepted measure to assess the discriminant validity (Wu & Yu, 2022). According to the criterion, discriminant validity could be reported as satisfactory when the correlation coefficients between variables are less than the square roots of AVE (Fornell & Larcker, 1981).

Table 2 shows the values of reliability, convergent validity, and discriminant validity in this study. All CR values range from 0.777 to 0.877, exceeding the threshold value of 0.07. Cronbach's alpha values range from 0.706 to 0.860. Besides, the item-total Cronbach's alpha is 0.933. These scales could reach a satisfactory level of reliability. All factor loadings in the study were found to be above the recommended threshold of 0.50 (Fornell & Larcker, 1981; Tang et al., 2021). Additionally, the Average Variance Extracted (AVE) values ranged from 0.470 to 0.601. It is worth noting that while some researchers argue that a Composite Reliability (CR) value above 0.6 and an AVE value higher than 0.4 indicate sufficient convergent validity (Fornell & Larcker, 1981; Tang et al., 2021). The convergent validity could be still considered acceptable. As shown in Table 3, the correlation coefficients between variables are less than all square roots of AVE, thus revealing an adequate level of discriminant validity.

Table 3 shows the means, standard deviations, and correlations of variables. This study analyzed the relationships between different variables using Pearson correlation coefficients. This study found that enjoyment is highly significantly associated with engagement ($r = 0.57$, $p < 0.01$). Adaptability also has a significant association with engagement ($r = 0.35$, $p < 0.01$). Similarly, extroversion, agreeableness, conscientiousness, neuroticism, and openness to a new experience were found to be significantly associated with engagement ($r = 0.14$, 0.36, 0.22, 12., 0.27 respectively, $p < 0.05$). In contrast to our expectation, anxiety has a weak and negative correlation to e-learning engagement ($r = -0.04$).

As can be seen from Table 4, the measurement model has a satisfactory fit to the data ($\chi 2 = 1768.33$; $\chi 2/d = 2.43$; SRMR = 0.049; CFI = 0.919; GFI = 0.919; TLI = 0.924; AGFI = 0.904; and RMSEA = 0.038). In addition, the observed variables could load on the corresponding latent variables ($p < 0.05$), suggesting that the observed variables could represent the corresponding latent variables.

5.1.2 Test of the structural equation modeling

The present study followed Preacher and Hayes' (2008) multiple mediation approach to test the mediating effects. Before testing the mediating model, this study initially developed a direct effect model to test the direct effect of personality on e-learning engagement. Looking at Table 4, the fit indices of the model are acceptable ($\chi 2 = 707.55$, $\chi 2/d = 3.41$, SRMR = 0.053, CFI = 0.928, GFI = 0.934, TLI = 0.917, AGFI = 0.918, and RMSEA = 0.049; Hu & Bentler, 1999).

As can be seen from Table 5, there are significant positive relationships between extroversion and e-learning engagement (estimate = 0.072, S.E. = 0.025, BC 95% CI = 0.029 to 0.117), agreeableness and e-learning engagement (estimate = 0.40, S.E. = 0.057, BC 95% CI = 0.294 to 0.526), conscientiousness and e-learning engagement (estimate = 0.179, S.E. = 0.03, BC 95% CI = 0.115 to 0.247), and openness to a new experience and e-learning engagement

Table 3 Means, standard deviations, and discriminant validity

Constructs	Mean	SD	1	2	3	4	5	6	7	8	9
1. EX	3.75	0.63	**0.68**								
2. AGR	3.51	0.68	0.12**	**0.61**							
3. CON	3.10	0.88	0.11**	0.08*	**0.67**						
4. NEURO	3.79	0.64	0.23**	0.01	0.38**	**0.71**					
5. OPEN	2.79	0.80	0.30**	0.27**	−0.04	−0.04	**0.74**				
6. ANX	3.73	0.58	−0.11**	−0.02	−0.27**	0.46**	−0.02	**0.69**			
7. EN	2.79	0.79	0.24**	0.31**	0.26**	−0.17**	0.33**	−0.10**	**0.66**		
8. AD	2.69	0.83	0.20**	0.29**	0.14**	0.31**	0.28**	0.16**	0.36**	**0.63**	
9. E	3.68	0.73	0.14**	0.36***	0.22**	0.12*	0.27**	−0.04	0.57**	0.35**	**0.72**

NEURO neuroticism; *EX* extroversion; *OPEN* openness to a new experience; *AGR* agreeableness; *CON* consciousness; *E* engagement; *ANX* anxiety; *AD* adaptability; *EN* enjoyment; *S.D.* standard deviation. Bold values represent the square roots of AVE. * $p < 0.05$, ** $p < 0.01$

Table 4 Model fit indices

Fit index	Measurement model	Direct effect Model	Multiple-mediation model	Recommended value
χ^2	1768.33	707.55	1778.40	
χ^2/df	2.43	3.41	2.43	< 5
RMSEA	0.038	0.049	0.038	< 0.05
CFI	0.933	0.928	0.932	> 0.09
TLI	0.924	0.917	0.924	> 0.09
SRMR	0.048	0.053	0.048	< 0.05
GFI	0.919	0.934	0.918	> 0.09
AGFI	0.904	0.918	0.903	> 0.09

(estimate = 0.079, S.E. = 0.032, BC 95% CI = 0.021 to 0.147). By contrast, there is a significant negative relationship between neuroticism and e-learning engagement (estimate = -0.099, S.E. = 0.039, BC 95% CI = -0.185 to -0.022). We thus accepted H1, H2, H3, H4, and H5.

This study tested a multiple-mediation model with enjoyment, anxiety, and adaptability as the mediators. The fit indices of the multiple-mediation model include $\chi 2 = 1952.56$, $\chi 2/d = 2.64$, SRMR = 0.048, CFI = 0.932, GFI = 0.918, TLI = 0.913, AGFI = 0.903, and RMSEA = 0.038. The multiple-mediation model also fits the data well (Hu & Bentler, 1999) (Fig. 2).

The results of bias-corrected bootstrap tests are shown in Table 5. All total effects are significant. Besides, the majority of direct effects are significant with the exception of the direct effect of extroversion on e-learning engagement (estimate = -0.008, S.E. = 0.027, BC 95% CI = -0.053 to 0.037) and openness to a new experience on e-learning engagement (estimate = 0.072, S.E. = 0.025, BC 95% CI = 0.029 to 0.117). Moreover, all indirect effects are significant with the exception of the indirect effect of neuroticism on e-learning engagement. Therefore, neuroticism could not exert an indirect effect on e-learning engagement via enjoyment, anxiety, and adaptability (estimate = 0.071, S.E. = 0.048, BC 95% CI = -0.01 to 0.147). We rejected H6d, H7d, and H8d.

Table 5 represents the summary statistics for the specific indirect effect. Extroversion, agreeableness, and openness to a new experience had an indirect effect on e-learning engagement via enjoyment and adaptability. In addition, conscientiousness had an indirect effect on e-learning engagement only through enjoyment. Anxiety, however, could not act as a mediator between personality and e-learning engagement. Therefore, we accepted H6a, H6b, H6c, H6e, H8a, H8c, and H8e but rejected H7a, H7b, H7c, H7e, and H8b.

Table 5 The SEM results (standardized coefficients)

Direct effect model				
Path	Points estimate	S. E	Bootstrapping BC 95% CI	
			Lower	upper
EX → E	0.072	0.025	0.029	0.117
AGR → E	0.40	0.057	0.294	0.526
CON → E	0.179	0.03	0.115	0.247
NEURO → E	−0.099	0.039	−0.185	−0.022
OPEN → E	0.079	0.032	0.021	0.147
Multiple-mediation model				
Total effect path	Points estimate	S. E	Bootstrapping BC 95% CI	
			Lower	upper
EX → E	0.066	0.029	0.022	0.115
AGR → E	0.488	0.081	0.374	0.638
CON → E	0.28	0.051	0.202	0.366
NEURO → E	−0.142	0.054	−0.235	−0.056
OPEN → E	0.091	0.041	0.028	0.164
Direct effect	Points estimate	S. E	Bootstrapping BC 95% CI	
			Lower	upper
EX → E	−0.008	0.027	−0.053	0.037
EX → EN	0.028	0.008	0.017	0.109
EX → ANX	−0.065	0.037	−0.128	−0.005
EX → AD	0.038	0.018	0.011	0.072
AGR → E	0.261	0.077	0.147	0.397
AGR → EN	0.074	0.022	0.043	0.109
AGR → ANX	−0.001	0.082	−0.134	0.142
AGR → AD	0.224	0.041	0.161	0.293
CON → E	0.153	0.045	0.079	0.226
CON → EN	0.051	0.014	0.032	0.109
CON → ANX	0.084	0.066	−0.016	0.199
CON → AD	0.002	0.026	−0.037	0.047
NEURO → E	−0.213	0.064	−0.31	−0.104
NEURO → EN	0.002	0.011	−0.015	0.109
NEURO → ANX	0.706	0.078	0.587	0.844
NEURO → AD	0.178	0.029	0.135	0.231
OPEN → E	0.02	0.036	−0.035	0.083
OPEN → EN	0.023	0.01	0.01	0.109
OPEN → ANX	0.009	0.051	−0.075	0.089
OPEN → AD	0.066	0.025	0.024	0.109
EN → E	0.436	0.628	0.514	0.738
ANX → E	0.042	0.042	−0.024	0.113
AD → E	0.21	0.117	0.016	0.396
Indirect effect Path	Points estimate	S. E	Bootstrapping BC 95% CI	
			Lower	upper
EX → E	0.074	0.018	0.047	0.106

Table 5 (continued)

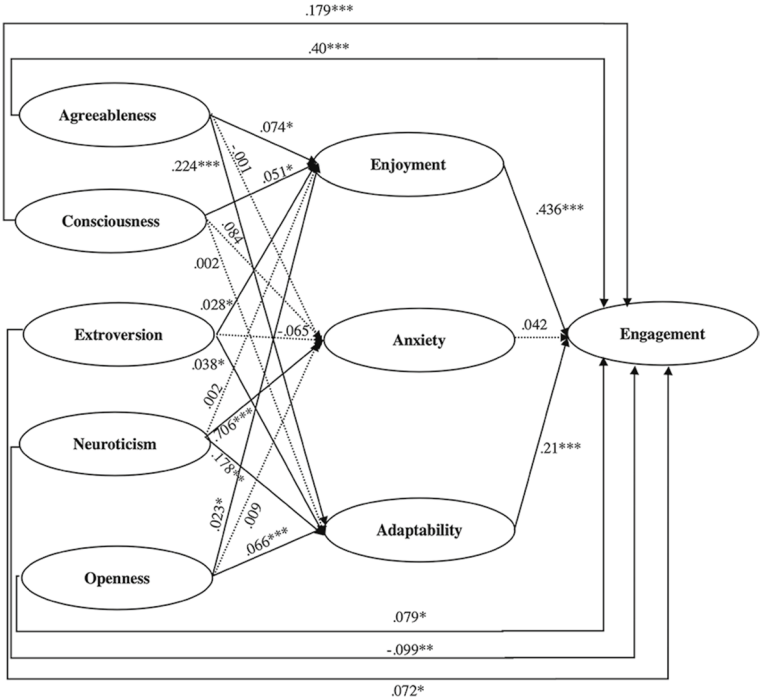
AGR → E	0.226	0.046	0.156	0.305
CON → E	0.128	0.03	0.084	0.182
NEURO → E	0.071	0.048	−0.01	0.147
OPEN → E	0.071	0.023	0.035	0.112
Specific indirect effect Path	Points estimate	S. E	Bootstrapping BC 95% CI	
			Lower	upper
EX → EN → E	0.068	0.016	0.044	0.098
EX → ANX → E	−0.003	0.003	−0.011	0.001
EX → AD → E	0.008	0.006	0.001	0.024
AGR → EN → E	0.179	0.038	0.126	0.249
AGR → ANX → E	0	0.005	−0.008	0.007
AGR → AD → E	0.047	0.026	0.006	0.091
CON → EN → E	0.124	0.027	0.083	0.17
CON → ANX → E	0.004	0.005	−0.001	0.017
CON → AD → E	0.001	0.006	−0.009	0.012
NEURO → EN → E	0.004	0.026	−0.037	0.047
NEURO → ANX → E	0.029	0.003	−0.015	0.081
NEURO → AD → E	0.037	0.022	0.005	0.077
OPEN → EN → E	0.057	0.019	0.026	0.09
OPEN → ANX → E	0	0.003	−0.003	0.007
OPEN → AD → E	0.014	0.009	0.002	0.034

NEURO neuroticism; *EX* extroversion; *OPEN* openness to a new experience; *AGR* agreeableness; *CON* consciousness; *E* engagement; *ANX* anxiety; *AD* adaptability; *EN* enjoyment; *S.E* Standard error; *BC* bias-corrected; *CI* confidence interval; 5,000 bootstrap samples

5.2 Results from the interviews

The qualitative findings from the interviews corroborate the results obtained from the questionnaire. The interviews revealed a strong association between personality traits and engagement in e-learning. It was observed that agreeable and extroverted students exhibited a high level of active involvement in e-learning. One interviewee who identified as agreeable expressed, "I feel engaged in e-learning as I cooperate with my classmates in a more flexible and diverse way." Similarly, another interviewee shared a similar experience, stating, "I feel involved in e-learning as I could express my ideas to my teachers in a timely manner." Notably, this interviewee, as per the measurement with the Big Five Inventory, exhibited an extroverted personality trait.

Furthermore, students with a strong level of conscientiousness reported a high extent of engagement in e-learning. An interviewee with pronounced conscientiousness mentioned, "Well, I could plan my work properly. I do not need to take part in other non-academic activities. I could completely engage in e-learning." Similarly, interviewees characterized by openness to new experiences displayed a preference for e-learning and demonstrated a deep sense of immersion in the



Note. The dashed lines represent that the predictive paths were nonsignificant. The bold lines represent that the predictive paths were significant. * $p < 0.05$, ** $p < 0.01$,*** $p < 0.001$

Fig. 2 SEM results

online learning environment. This finding is supported by one interviewee's statement, declaring, "I find myself more motivated to engage in e-learning." However, students with high levels of neuroticism reported lower levels of engagement in e-learning. One interviewee with high neuroticism candidly admitted, "To be honest, I find it difficult to engage in e-learning due to the prolonged screen exposure" (Mikulincer et al., 2003).

The qualitative findings also provide evidence that the achievement emotion of enjoyment may act as a mediator between personality traits and e-learning engagement. For example, one agreeable interviewee expressed, "I feel joyful and engaged in e-learning because I could stay connected with peers and teachers easily, anywhere, and anytime." Similarly, an extroverted interviewee highlighted the role of enjoyment, stating, "Well, my teachers often post interesting materials on educational apps. I like to discuss these interesting materials with my peers in online classes. It is fun and engaging" (Fredrickson, 2001; Rozin, 1999).

Moreover, an interviewee with a strong openness to new experiences emphasized the enjoyment derived from e-learning, stating, "I am curious and ready to learn new knowledge. Attending online courses makes it easy for us to learn various knowledge. It makes me feel enjoyable and engaged in e-learning" (John & Srivastava,

1999). Conversely, the majority of interviewees did not perceive anxiety as having a profound impact on their e-learning engagement.

Additionally, the results from the interviews revealed that adaptability may mediate the relationship between personality traits and e-learning engagement. An interviewee with a strong openness to new experiences noted, "I am supposed to be adaptable. I like to use various new educational technologies. Using new technologies could help me adjust and stay focused in online classes" (McCrae & Costa, 1999). Similarly, an extroverted interviewee stated, "I am outgoing and well-adjusted. I am soon deeply involved in e-learning during the pandemic because I have the ability to cope with difficulties and adapt to e-learning".

In summary, the qualitative results provide further support for the mediating role of enjoyment and adaptability in the relationship between personality traits and e-learning engagement. Enjoyment derived from e-learning experiences was found to be influenced by agreeableness, extroversion, and openness to new experiences, while adaptability was associated with openness to new experiences and extroversion. These findings contribute to our understanding of the underlying mechanisms influencing e-learning engagement among students (Carver & Scheier, 2002; Watson & Hubbard, 1996).

6 Discussion

The present study provides evidence of a significant relationship between personality traits and e-learning engagement. Consistent with previous research findings (Audet et al., 2021; Quigley et al., 2022), agreeableness, conscientiousness, extroversion, and openness to a new experience were positively associated with higher levels of e-learning engagement. Conversely, individuals high in neuroticism showed lower levels of e-learning engagement. These findings indicate that teachers need to consider the diverse needs and preferences of students in the e-learning environment.

Furthermore, the results suggest that students with higher scores on extroversion, agreeableness, conscientiousness, and openness to a new experience experienced enjoyment in their e-learning experiences, which subsequently increased their levels of e-learning engagement. Extroverted students derived enjoyment from engaging in various activities that involved communication with others in the e-learning context (Hong et al., 2021). Agreeable students found enjoyment in developing positive teacher-student relationships during e-learning. Conscientious students experienced enjoyment as they successfully completed their learning tasks. Open-minded students derived enjoyment from the novel learning experiences presented in e-learning environments. This enjoyment, in turn, fostered confidence and comfort in these students (Sadoughi & Hejazi, 2021).

Moreover, enjoyment was found to facilitate students' motivation, creative breakthroughs, and persistence, thereby enhancing their engagement in e-learning (Sadoughi & Hejazi, 2021). By experiencing enjoyment in their e-learning activities, students were more likely to be motivated to participate actively, think critically, and contribute creatively to their learning process.

To sum up, the findings of this study underscore the importance of considering personality traits in understanding e-learning engagement. The results highlight the positive association between personality traits (such as agreeableness, conscientiousness, extroversion, and openness to a new experience) with e-learning engagement. Additionally, enjoyment acted as a mediator between personality traits and e-learning engagement, with extroversion, agreeableness, conscientiousness, and openness to a new experience influencing enjoyment in e-learning experiences. These findings provide valuable insights for educators, emphasizing the significance of catering to individual differences in personality and fostering enjoyment to enhance students' engagement in e-learning.

It was surprising to find that enjoyment did not mediate the relationship between neuroticism and e-learning engagement. One possible explanation for this finding is that neuroticism is more strongly related to negative emotions rather than positive emotions (de la Fuente et al., 2020; Hong et al., 2021). In other words, neuroticism may not predict enjoyment as effectively as other personality traits. This could explain why highly neurotic students were less likely to experience enjoyment in the e-learning context.

Furthermore, an interesting result from the data analysis was the lack of a significant mediation effect of anxiety in the relationship between personality and e-learning engagement. Specifically, there was no significant correlation found between personality traits and anxiety levels. Additionally, no potential association was observed between anxiety and e-learning engagement. These findings are in line with previous research (Pekrun, 2006; Wang et al., 2022). These results may be attributed to the complex nature of anxiety. While anxiety has been shown to have a negative impact on students' intrinsic interest and engagement in online courses, it may also increase extrinsic motivation to avoid failure, thereby promoting engagement in e-learning (Liu et al., 2021a, b).

The mediating role of adaptability in the relationship between personality and e-learning engagement is supported by the findings of Martos Martínez et al. (2021). Students with higher levels of agreeableness, extroversion, and openness to new experiences tend to be more adaptable and show greater tolerance for uncertainty, leading to more positive reactions to e-learning (Besser et al., 2022; Varela et al., 2012; Zhang et al., 2021). This adaptability, in turn, increases their interest in e-learning and enhances their engagement (Martin & Borup, 2022), aligning with the ideas proposed by Martin et al. (2013) regarding the mediating role of adaptability in the association between personality and engagement.

Contrary to expectations, however, adaptability did not act as a mediator between conscientiousness and e-learning engagement. This result contradicts the findings of Martin et al. (2013) who observed the mediating role of adaptability in the effect of conscientiousness on learning engagement. This unexpected result may be attributed to multiple mediators in our model suppressing each other. Specifically, the inclusion of enjoyment as a mediator weakened the effect of conscientiousness on adaptability. This finding is supported by the observation that the indirect effect through enjoyment had a stronger influence than the indirect effect through adaptability. Nonetheless, this surprising finding suggests that enjoyment may enable highly conscientious students to have a higher level of e-learning engagement compared to adaptability alone.

7 Conclusion

7.1 Major findings

This present study employed the five-factor model of personality to explore the relationships between university students' personality traits and e-learning engagement and the mediating effects of achievement emotions and adaptability The findings showed that there was a significant relationship between personality and e-learning engagement. Additionally, this study found that enjoyment could act as a mediator between personality traits (excluding neuroticism) and e-learning engagement. However, anxiety could not play a mediating role in the relationship between personality and e-learning engagement. Moreover, the results revealed that in the e-learning context, adaptability could be a mediator between personality traits (excluding conscientiousness and neuroticism) and e-learning engagement.

7.2 Theoretical contributions

The findings from this study make contributions to the current literature. Initially, this study responded to a call for more investigation of the relationship between student personality and engagement in the e-learning context (Quigley et al., 2022). This study advances the understanding of the direct roles of personality traits in e-learning engagement. More importantly, this study shed light on the indirect roles of personality traits in e-learning engagement. This study takes a more comprehensive view of the personality-engagement relationship in the emerging field of e-learning.

Second, to our knowledge, little published research has examined whether achievement emotions and adaptability could have mediating effects on the relationship between personality and engagement in the e-learning context. This study makes a major contribution by revealing that enjoyment and adaptability could mediate the relationship between personality traits and e-learning engagement, whereas anxiety could not mediate this relationship. We highlighted and confirmed the importance of enjoyment and adaptability in the e-learning environment. The findings of the current study reinforced the idea that enjoyment and adaptability could enable university students to have a higher level of e-learning engagement.

7.3 Practical implications

The findings of this study have practical implications for researchers and instructors who wish to improve students' engagement in e-learning. First, due to the role of personality in e-learning engagement, there is an urgent need to tailor learning content and activities to satisfy the diverse needs of students with different personality traits. Emerging educational computer programs could be used to provide personalized learning. For example, ChatGPT with characteristics of interactivity, flexibility, and personalization could provide personalized learning content and feedback to

students with different personalities, enabling them to have a great learning experience and engagement in e-learning (Al-Sharafi et al., 2022).

Moreover, the findings shed new light on how students' achievement emotions and adaptability could mediate the personality-engagement relationship, particularly in the e-learning context. This suggests that boosting students' positive achievement emotions and adaptability could be conducive to students' learning engagement. It highlights the need for researchers and practitioners to intervene with some personality types who have low levels of enjoyment and high levels of anxiety in e-learning. Researchers and instructors can use educational games to increase these students' enjoyment and reduce their anxiety in the learning process. Researchers and instructors could also integrate warning systems into e-learning, as this could help them to periodically detect and intervene with certain personality types who require additional support in effectively participating in e-learning.

7.4 Limitations

Some limitations need to be noted regarding our study. Firstly, given the self-report nature of the online questionnaire, recall bias and social desirability bias may occur in this study when some students might give desirable answers to avoid embarrassment and maintain positive self-images. Second, some caution must be applied regarding the generalizability of these findings. Due to resource and time constraints, the present study randomly selected students from Guizhou University. The findings might thus not be generalizable to other universities. Additionally, this study is limited by an unequal gender ratio in the sample, with a higher proportion of females than males. Thirdly, the scope of this study might be relatively narrow. There might be reciprocal relationships among these variables, such as achievement emotions and learning engagement. However, this study only focused on the effects of achievement emotions on learning engagement.

7.5 Suggestions for future studies

Understanding how and why student personality influences e-learning engagement is crucial for educational practitioners to enhance teaching quality and foster online learners' engagement. Exploring the underlying mechanism between personality traits and different forms of e-learning engagement holds great promise. Subsequent studies could delve into comparing various types of online courses, such as MOOCs and video conferencing, and investigate how specific features of these courses mediate the relationship between personality and e-learning engagement. Additionally, psychological factors, such as students' prior knowledge, flow, procrastination, and perceptions, may significantly impact the personality-engagement relationship within e-learning environments. Further research is needed to ascertain the significance of these factors in complex relationships. Moreover, future studies on this subject should consider incorporating alternative data analysis methods, such as cluster analysis, to enhance our comprehension of the personality-engagement relationship in the context of e-learning.

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Authors' contributions Rong Wu: Methodology, Investigation, Editing, and Writing-Original Draft. Zhonggen Yu: Conceptualization and Funding acquisition.

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Data availability The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Declarations

Ethics approval statement The study was approved by the institutional review board of Beijing Language and Culture University. All researchers can provide written informed consents.

Informed consent All participants voluntarily signed the consent form before answering the questionnaire.

Conflict of interest/competing interests We have no conflicts of interest to declare that are relevant to the content of this article.

References

Adeshola, I., & Agoyi, M. (2022). Examining factors influencing e-learning engagement among university students during covid-19 pandemic: A mediating role of "learning persistence". *Interactive Learning Environments,* 1–28. https://doi.org/10.1080/10494820.2022.2029493

Alon, L., Sung, S., Cho, J., & Kizilcec, R. F. (2023). From emergency to sustainable e-learning: Changes and disparities in undergraduate course grades and experiences in the context of COVID-19. *Computers & Education, 203*, 104870. https://doi.org/10.1016/j.compedu.2023.104870

Al-Sharafi, M. A., Al-Emran, M., Iranmanesh, M., Al-Qaysi, N., Iahad, N. A., & Arpaci, I. (2022). Understanding the impact of knowledge management factors on the sustainable use of AI-based chatbots for educational purposes using a hybrid SEM-ANN approach. *Interactive Learning Environments,* 1–21. https://doi.org/10.1080/10494820.2022.2075014

Anderson, J. C., & Gerbing, D. W. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin, 103*, 411–423. https://doi.org/10.1037/0033-2909.103.3.411

Archambault, L., Leary, H., & Rice, K. (2022). Pillars of online pedagogy: A framework for teaching in e-learning environments. *Educational Psychologist, 57*(3), 178–191. https://doi.org/10.1080/00461520.2022.2051513

Audet, É. C., Levine, S. L., Metin, E., Koestner, S., & Barcan, S. (2021). Zooming their way through university: Which Big 5 traits facilitated students' adjustment to online courses during the COVID-19 pandemic. *Personality and Individual Differences, 180*, 110969. https://doi.org/10.1016/j.paid.2021.110969

Baruth, O., & Cohen, A. (2023). Personality and satisfaction with online courses: The relation between the Big Five personality traits and satisfaction with e-learning activities. *Education and Information Technologies, 28*(1), 879–904. https://doi.org/10.1007/s10639-022-11199-x

Bentler, P. M., & Bonett, D. G. (1980). Significance tests and goodness of fit in the analysis of covariance structures. *Psychological Bulletin, 88*, 588–606. https://doi.org/10.1037/0033-2909.88.3.588

Besser, A., Flett, G. L., & Zeigler-Hill, V. (2022). Adaptability to a sudden transition to e-learning during the COVID-19 pandemic: Understanding the challenges for students. *Scholarship of Teaching and Learning in Psychology, 8*(2), 85–105. https://doi.org/10.1037/stl0000198

Bhagat, K. K., Wu, L. Y., & Chang, C.-Y. (2019). The impact of personality on students' perceptions towards online learning. *Australasian Journal of Educational Technology, 35*(4), 4. https://doi.org/10.14742/ajet.4162

Bosselut, G., Castro, O., Chevalier, S., & Fouquereau, E. (2020). Does perceived cohesion mediate the student personality–engagement relationship in the university setting? *Journal of Educational Psychology, 112*(8), 1692–1700. https://doi.org/10.1037/edu0000442

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Carver, C. S., & Scheier, M. F. (2002). Control processes and self-organization as complementary principles underlying behavior. *Personality and Social Psychology Review, 6*(4), 304–315. https://doi.org/10.1207/S15327957PSPR0604_05

Chen, P., Bao, C., & Gao, Q. (2021). Proactive personality and academic engagement: The mediating effects of teacher-student relationships and academic self-efficacy. *Frontiers in Psychology, 12*, 652994. https://doi.org/10.3389/fpsyg.2021.652994

Chen, X., He, J., Swanson, E., Cai, Z., & Fan, X. (2022). Big five personality traits and second language learning: A meta-analysis of 40 years' research. *Educational Psychology Review, 34*(2), 851–887. https://doi.org/10.1007/s10648-021-09641-6

Closson, L. M., & Boutilier, R. R. (2017). Perfectionism, academic engagement, and procrastination among undergraduates: The moderating role of honors student status. *Learning and Individual Differences, 57*, 157–162. https://doi.org/10.1016/j.lindif.2017.04.010

Collie, R. J., & Martin, A. J. (2017). Students' adaptability in mathematics: Examining self-reports and teachers' reports and links with engagement and achievement outcomes. *Contemporary Educational Psychology, 49*, 355–366. https://doi.org/10.1016/j.cedpsych.2017.04.001

Collie, R. J., Holliman, A. J., & Martin, A. J. (2017). Adaptability, engagement and academic achievement at university. *Educational Psychology, 37*(5), 632–647. https://doi.org/10.1080/01443410.2016.1231296

Collie, R. J., Granziera, H., & Martin, A. J. (2018). Teachers' perceived autonomy support and adaptability: An investigation employing the job demands-resources model as relevant to workplace exhaustion, disengagement, and commitment. *Teaching and Teacher Education, 74*, 125–136. https://doi.org/10.1016/j.tate.2018.04.015

Creswell, J. W., & Clark, V. L. P. (2017). *Designing and Conducting Mixed Methods Research*. SAGE Publications.

de la Fuente, J., Paoloni, P., Kauffman, D., Yilmaz Soylu, M., Sander, P., & Zapata, L. (2020). Big Five, self-regulation, and coping strategies as predictors of achievement emotions in undergraduate students. *International Journal of Environmental Research and Public Health, 17*(10), 10. https://doi.org/10.3390/ijerph17103602

Deng, W., Lei, W., Guo, X., Li, X., Ge, W., & Hu, W. (2022). Effects of regulatory focus on e-learning engagement of high school students: The mediating role of self-efficacy and academic emotions. *Journal of Computer Assisted Learning, 38*(3), 707–718. https://doi.org/10.1111/jcal.12642

Dewaele, J. M., & Li, C. (2021). Teacher enthusiasm and students' social-behavioral learning engagement: The mediating role of student enjoyment and boredom in Chinese EFL classes. *Language Teaching Research, 25*(6), 922–945. https://doi.org/10.1177/13621688211014538

Earl, S. R., Taylor, I. M., Meijen, C., & Passfield, L. (2023). Trajectories in cognitive engagement, fatigue, and school achievement: The role of young adolescents' psychological need satisfaction. *Learning and Individual Differences, 101*, 102248. https://doi.org/10.1016/j.lindif.2022.102248

Ewing, L. A., & Cooper, H. B. (2021). Technology-enabled remote learning during Covid-19: Perspectives of Australian teachers, students and parents. *Technology, Pedagogy and Education, 30*(1), 41–57. https://doi.org/10.1080/1475939X.2020.1868562

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research, 18*(1), 39–50. https://doi.org/10.1177/002224378101800104

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research, 74*(1), 59–109. https://doi.org/10.3102/00346543074001059

Fredrickson, B. L. (2001). The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions. *American Psychologist, 56*, 218–226. https://doi.org/10.1037/0003-066X.56.3.218

Goldberg, L. R. (1993). The structure of phenotypic personality traits. *American Psychologist, 48*(1), 26–34. https://doi.org/10.1037/0003-066X.48.1.26

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2014). *Multivariate data analysis: A global perspective*. Pearson Education.

Holliman, A. J., Martin, A. J., & Collie, R. J. (2018). Adaptability, engagement, and degree completion: A longitudinal investigation of university students. *Educational Psychology, 38*(6), 785–799. https://doi.org/10.1080/01443410.2018.1426835

Hong, J., Cao, W., Liu, X., Tai, K., & Zhao, L. (2021). Personality traits predict the effects of Internet and academic self-efficacy on practical performance anxiety in e-learning under the COVID-19 lockdown. *Journal of Research on Technology in Education, 55*(3), 426–440. https://doi.org/10.1080/15391523.2021.1967818

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal, 6*(1), 1–55. https://doi.org/10.1080/10705519909540118

Iterbeke, K., & De Witte, K. (2021). Helpful or harmful? The role of personality traits in student experiences of the COVID-19 crisis and school closure. *Personality and Social Psychology Bulletin, 48*(11), 1614–1632. https://doi.org/10.1177/01461672211050515

Izhar, L. I., Babiker, A., Rizki, E. E., Lu, C.-K., & Abdul Rahman, M. (2022). Emotion self-regulation in neurotic students: A pilot mindfulness-based intervention to assess its effectiveness through brain signals and behavioral data. *Sensors, 22*(7), 7. https://doi.org/10.3390/s22072703

John, O. P., & Srivastava, S. (1999). The big five trait taxonomy: history, measurement, and theoretical perspectives. *In Handbook of Personality: Theory and Research* (pp. 102–138). Guilford Press

Jung, Y., & Lee, J. (2018). Learning engagement and persistence in Massive Open Online Courses (MOOCS). *Computers & Education, 122*, 9–22. https://doi.org/10.1016/j.compedu.2018.02.013

Keller, H., & Karau, S. J. (2013). The importance of personality in students' perceptions of the e-learning experience. *Computers in Human Behavior, 29*(6), 2494–2500. https://doi.org/10.1016/j.chb.2013.06.007

Komarraju, M., & Karau, S. J. (2005). The relationship between the big five personality traits and academic motivation. *Personality and Individual Differences, 39*(3), 557–567. https://doi.org/10.1016/j.paid.2005.02.013

Lee, P., & Wu, T. (2022). Mining relations between personality traits and learning styles. *Information Processing & Management, 59*(5), 103045. https://doi.org/10.1016/j.ipm.2022.103045

Li, F., Jin, T., Edirisingha, P., & Zhang, X. (2021). School-aged students' sustainable e-learning engagement during COVID-19: Community of inquiry in a Chinese secondary education context. *Sustainability, 13*(18), 18. https://doi.org/10.3390/su131810147

Liu, H., Yao, M., Li, J., & Li, R. (2021a). Multiple mediators in the relationship between perceived teacher autonomy support and student engagement in math and literacy learning. *Educational Psychology, 41*(2), 116–136. https://doi.org/10.1080/01443410.2020.1837346

Liu, X., Gong, S., Zhang, H., Yu, Q., & Zhou, Z. (2021b). Perceived teacher support and creative self-efficacy: The mediating roles of autonomous motivation and achievement emotions in Chinese junior high school students. *Thinking Skills and Creativity, 39*, 100752. https://doi.org/10.1016/j.tsc.2020.100752

Liu, Y., Zhang, M., Qi, D., & Zhang, Y. (2022). Understanding the role of learner engagement in determining MOOCs satisfaction: A self-determination theory perspective. *Interactive Learning Environments,* 1–15. https://doi.org/10.1080/10494820.2022.2028853

Martin, F., & Borup, J. (2022). Online learner engagement: Conceptual definitions, research themes, and supportive practices. *Educational Psychologist, 57*(3), 162–177. https://doi.org/10.1080/00461520.2022.2089147

Martin, A. J., Nejad, H., Colmar, S., & Liem, G. A. D. (2012). Adaptability: Conceptual and empirical perspectives on responses to change, novelty and uncertainty. *Journal of Psychologists and Counsellors in Schools, 22*(1), 58–81. https://doi.org/10.1017/jgc.2012.8

Martin, A. J., Nejad, H. G., Colmar, S., & Liem, G. A. D. (2013). Adaptability: How students' responses to uncertainty and novelty predict their academic and non-academic outcomes. *Journal of Educational Psychology, 105*(3), 728–746. https://doi.org/10.1037/a0032794

Martin, A. J., Yu, K., Ginns, P., & Papworth, B. (2017). Young people's academic buoyancy and adaptability: A cross-cultural comparison of China with North America and the United Kingdom. *Educational Psychology, 37*(8), 930–946. https://doi.org/10.1080/01443410.2016.1202904

Martos Martínez, Á., Pérez-Fuentes, M. D. C., Molero Jurado, M. D. M., Simón Márquez, M. D. M., Barragán Martín, A. B., & Gázquez Linares, J. J. (2021). Empathy, affect and personality as predictors of engagement in nursing professionals. *International Journal of Environmental Research and Public Health, 18*(8), 8. https://doi.org/10.3390/ijerph18084110

McCrae, R. R., & Costa, P. T. (1987). Validation of the five-factor model of personality across instruments and observers. *Journal of Personality and Social Psychology, 52*(1), 81–90. https://doi.org/10.1037/0022-3514.52.1.81

McCrae, R. R., & John, O. P. (1992). An introduction to the five-factor model and its applications. *Journal of Personality, 60*(2), 175–215. https://doi.org/10.1111/j.1467-6494.1992.tb00970.x

McCrae, R. R., & Costa, P. T., Jr. (1999). A Five-Factor theory of personality. In L. A. Pervin & O. P. John (Eds.), *Handbook of personality: Theory and research* (pp. 139–153). Guilford Press.

Meyer, D. K., & Turner, J. C. (2002). Using instructional discourse analysis to study the scaffolding of student self-regulation. *Educational Psychologist, 37*(1), 17–25. https://doi.org/10.1207/S15326985EP3701_3

Mihai, M., Albert, C. N., Mihai, V. C., & Dumitras, D. E. (2022). Emotional and social engagement in the English language classroom for higher education students in the COVID-19 online context. *Sustainability, 14*(8), 8. https://doi.org/10.3390/su14084527

Mikulincer, M., Shaver, P. R., & Pereg, D. (2003). Attachment theory and affect regulation: The dynamics, development, and cognitive consequences of attachment-related strategies. *Motivation and Emotion, 27*(2), 77–102.

Moreira, P. A. S., Inman, R. A., Cloninger, K., & Cloninger, C. R. (2021). Student engagement with school and personality: A biopsychosocial and person-centred approach. *British Journal of Educational Psychology, 91*(2), 12388. https://doi.org/10.1111/bjep.12388

Nungu, L., Mukama, E., & Nsabayezu, E. (2023). Online collaborative learning and cognitive presence in mathematics and science education. Case study of university of Rwanda, college of education. *Education and Information Technologies, 28*, 10865–10884. https://doi.org/10.1007/s10639-023-11607-w

Pekrun, R. (2006). The control-value theory of achievement emotions: Assumptions, corollaries, and implications for educational research and practice. *Educational Psychology Review, 18*(4), 315–341. https://doi.org/10.1007/s10648-006-9029-9

Pekrun, R. (2017). Emotion and achievement during adolescence. *Child Development Perspectives, 11*(3), 215–221. https://doi.org/10.1111/cdep.12237

Pekrun, R., Goetz, T., Titz, W., & Perry, R. P. (2002). Academic emotions in students' self-regulated learning and achievement: A program of qualitative and quantitative research. *Educational Psychologist, 37*(2), 91–105. https://doi.org/10.1207/S15326985EP3702_4

Pekrun, R., Goetz, T., Frenzel, A. C., Barchfeld, P., & Perry, R. P. (2011). Measuring emotions in students' learning and performance: The Achievement Emotions Questionnaire (AEQ). *Contemporary Educational Psychology, 36*(1), 36–48. https://doi.org/10.1016/j.cedpsych.2010.10.002

Peng, X., Chen, H., Wang, L., Tian, F., & Wang, H. (2020). Talking head-based L2 pronunciation training: Impact on achievement emotions, cognitive load, and their relationships with learning performance. *International Journal of Human-Computer Interaction, 36*(16), 1487–1502. https://doi.org/10.1080/10447318.2020.1752476

Piaget, J. (1952). *The Origins of Intelligence in Children*. International Universities Press.

Poon, W. C., Kunchamboo, V., & Koay, K. Y. (2022). E-Learning enagagement and effectiveness during the COVID-19 Pandemic: The interaction model. *International Journal of Human–Computer Interaction,* 1–15. https://doi.org/10.1080/10447318.2022.2119659

Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods, 40*(3), 879–891. https://doi.org/10.3758/BRM.40.3.879

Quigley, M., Bradley, A., Playfoot, D., & Harrad, R. (2022). Personality traits and stress perception as predictors of students' online engagement during the COVID-19 pandemic. *Personality and Individual Differences, 194*, 111645. https://doi.org/10.1016/j.paid.2022.111645

Qureshi, A., Wall, H., Humphries, J., & BahramiBalani, A. (2016). Can personality traits modulate student engagement with learning and their attitude to employability? *Learning and Individual Differences, 51*, 349–358. https://doi.org/10.1016/j.lindif.2016.08.026

Reeve, J., & Tseng, C. M. (2011). Agency as a fourth aspect of students' engagement during learning activities. *Contemporary Educational Psychology, 36*(4), 257–267. https://doi.org/10.1016/j.cedpsych.2011.05.002

Rozin, P. (1999). The process of moralization. *Psychological Science, 10*(3), 218–221. https://doi.org/10.1111/1467-9280.00139

Sadoughi, M., & Hejazi, S. Y. (2021). Teacher support and academic engagement among EFL learners: The role of positive academic emotions. *Studies in Educational Evaluation, 70*, 101060. https://doi.org/10.1016/j.stueduc.2021.101060

Sak, M. (2021). Understanding the role of personality in explaining L2 learners' DMC disposition. *Foreign Language Annals, 54*(2), 429–451. https://doi.org/10.1111/flan.12524

Sander, P., & de la Fuente, J. (2022). Modelling students' academic confidence, personality and academic emotions. *Current Psychology, 41*(7), 4329–4340. https://doi.org/10.1007/s12144-020-00957-0

Santo, L. D., Peña-Jimenez, M., Canzan, F., Saiani, L., & Battistelli, A. (2022). The emotional side of the e-learning among nursing students: The role of the affective correlates on e-learning satisfaction. *Nurse Education Today, 110*, 105268. https://doi.org/10.1016/j.nedt.2022.105268

Saunders, B., Sim, J., Kingstone, T., Baker, S., Waterfield, J., Bartlam, B., Burroughs, H., & Jinks, C. (2018). Saturation in qualitative research: Exploring its conceptualization and operationalization. *Quality & Quantity, 52*(4), 1893–1907. https://doi.org/10.1007/s11135-017-0574-8

Slof, B., van Leeuwen, A., Janssen, J., & Kirschner, P. A. (2021). Mine, ours, and yours: Whose engagement and prior knowledge affects individual achievement from online collaborative learning? *Journal of Computer Assisted Learning, 37*(1), 39–50. https://doi.org/10.1111/jcal.12466

Spielmann, J., Yoon, H. J. R., Ayoub, M., Chen, Y., Eckland, N. S., Trautwein, U., Zheng, A., & Roberts, B. W. (2022). An in-depth review of conscientiousness and educational issues. *Educational Psychology Review, 34*, 2745–2781. https://doi.org/10.1007/s10648-022-09693-2

Sulea, C., van Beek, I., Sarbescu, P., Virga, D., & Schaufeli, W. B. (2015). Engagement, boredom, and burnout among students: Basic need satisfaction matters more than personality traits. *Learning and Individual Differences, 42*, 132–138. https://doi.org/10.1016/j.lindif.2015.08.018

Tang, Y. M., Chen, P. C., Law, K. M. Y., Wu, C. H., Lau, Y., Guan, J., He, D., & Ho, G. T. S. (2021). Comparative analysis of Student's live e-learning readiness during the coronavirus (COVID-19) pandemic in the higher education sector. *Computers & Education, 168*, 104211. https://doi.org/10.1016/j.compedu.2021.104211

Varela, O. E., Cater, J. J., & Michel, N. (2012). E-learning in management education: An empirical study of the role of personality traits. *Journal of Computing in Higher Education, 24*(3), 209–225. https://doi.org/10.1007/s12528-012-9059-x

Wang, Y., Cullen, K. L., Yao, X., & Li, Y. (2013). Personality, freshmen proactive social behavior, and college transition: Predictors beyond academic strategies. *Learning and Individual Differences, 23*, 205–212. https://doi.org/10.1016/j.lindif.2012.09.010

Wang, J., Liu, R., Ding, Y., Xu, L., Liu, Y., & Zhen, R. (2017). Teacher's autonomy support and engagement in Math: Multiple mediating roles of self-efficacy, intrinsic value, and boredom. *Frontiers in Psychology, 8*, 1006. https://doi.org/10.3389/fpsyg.2017.01006

Wang, Y., Cao, Y., Gong, S., Wang, Z., Li, N., & Ai, L. (2022). Interaction and learning engagement in e-learning: The mediating roles of e-learning self-efficacy and academic emotions. *Learning and Individual Differences, 94*, 102128. https://doi.org/10.1016/j.lindif.2022.102128

Wang, X., Liu, Y., Ying, B., & Lin, J. (2023). The effect of learning adaptability on Chinese middle school students' English academic engagement: The chain mediating roles of foreign language anxiety and English learning self-efficacy. *Current Psychology, 42*(8), 6682–6692. https://doi.org/10.1007/s12144-021-02008-8

Watson, D., & Hubbard, B. (1996). Adaptational style and dispositional structure: Coping in the context of the Five-Factor model. *Journal of Personality, 64*(4), 737–774. https://doi.org/10.1111/j.1467-6494.1996.tb00943.x

Wester, E. R., Walsh, L. L., Arango-Caro, S., & Callis-Duehl, K. L. (2021). Student engagement declines in STEM undergraduates during COVID-19–driven remote learning. *Journal of Microbiology & Biology Education, 22*(1), 1–11. https://doi.org/10.1128/jmbe.v22i1.2385

Wildermuth, C. D. M. E. S., Vaughan, A. G., & Christo-Baker, E. A. (2013). A path to passion: connecting personality, psychological conditions, and emotional engagement. *Journal of Psychological Issues in Organizational Culture, 3*(4), 18–45. https://doi.org/10.1002/jpoc.21082

Wilmot, M. P., & Ones, D. S. (2022). Agreeableness and its consequences: A quantitative review of meta-analytic findings. *Personality and Social Psychology Review, 26*(3), 242–280. https://doi.org/10.1177/10888683211073007

Woods, S. A., & Sofat, J. A. (2013). Personality and engagement at work: The mediating role of psychological meaningfulness. *Journal of Applied Social Psychology, 43*(11), 2203–2210. https://doi.org/10.1111/jasp.12171

Wu, F., & Teets, T. S. (2021). Effects of the COVID-19 pandemic on student engagement in a general chemistry course. *Journal of Chemical Education, 98*(12), 3633–3642. https://doi.org/10.1021/acs.jchemed.1c00665

Wu, C. H., Parker, S. K., Wu, L. Z., & Lee, C. (2018). When and why people engage in different forms of proactive behavior: Interactive effects of self-construals and work characteristics. *Academy of Management Journal, 61*(1), 293–323. https://doi.org/10.5465/amj.2013.1064

Wu, R., & Yu, Z. (2022). The influence of social isolation, technostress, and personality on the acceptance of online meeting platforms during the COVID-19 Pandemic. *International Journal of Human–Computer Interaction,* 1–18. https://doi.org/10.1080/10447318.2022.2097779

Yan, L., Whitelock-Wainwright, A., Guan, Q., Wen, G., Gašević, D., & Chen, G. (2021). Students' experience of e-learning during the COVID-19 pandemic: A province-wide survey study. *British Journal of Educational Technology, 52*(5), 2038–2057. https://doi.org/10.1111/bjet.13102

Yilmaz, M. B., Orhan, F., & Zeren, S. G. (2022). Adolescent emotion scale for online lessons: A study from Turkey. *Education and Information Technologies, 27*(3), 3403–3420. https://doi.org/10.1007/s10639-021-10734-6

Yu, Z. (2021). The effects of gender, educational level, and personality on e-learning outcomes during the COVID-19 pandemic. *International Journal of Educational Technology in Higher Education, 18*(1), 14. https://doi.org/10.1186/s41239-021-00252-3

Yu, Z. (2022). A meta-analysis and bibliographic review of the effect of nine factors on e-learning outcomes across the world. *Education and Information Technologies, 27*(2), 2457–2482. https://doi.org/10.1007/s10639-021-10720-y

Yu, Z., Xu, W., & Sukjairungwattana, P. (2022). A meta-analysis of eight factors influencing MOOC-based learning outcomes across the world. *Interactive Learning Environments,* 1–20. https://doi.org/10.1080/10494820.2022.2096641

Yusoff, M. S. B., Hadie, S. N. H., & Yasin, M. A. M. (2021). The roles of emotional intelligence, neuroticism, and academic stress on the relationship between psychological distress and burnout in medical students. *BMC Medical Education, 21*(1), 293. https://doi.org/10.1186/s12909-021-02733-5

Zhang, K., Wu, S., Xu, Y., Cao, W., Goetz, T., & Parks-Stamm, E. J. (2021). Adaptability promotes student engagement under COVID-19: The multiple mediating effects of academic emotion. *Frontiers in Psychology, 11*, 633265. https://doi.org/10.3389/fpsyg.2020.633265

Zhang, R., Bi, N. C., & Mercado, T. (2023). Do zoom meetings really help? A comparative analysis of synchronous and asynchronous e-learning during Covid-19 pandemic. *Journal of Computer Assisted Learning, 39*(1), 210–217. https://doi.org/10.1111/jcal.12740

Zhao, L., Cao, C., Li, Y., & Li, Y. (2022). Determinants of the digital outcome divide in E-learning between rural and urban students: Empirical evidence from the COVID-19 pandemic based on capital theory. *Computers in Human Behavior, 130*, 107177. https://doi.org/10.1016/j.chb.2021.107177

Zheng, Y., Yu, S., & Liu, Z. (2023). Understanding individual differences in lower-proficiency students' engagement with teacher written corrective feedback. *Teaching in Higher Education, 28*(2), 301–321. https://doi.org/10.1080/13562517.2020.1806225

Zyphur, M. J., Bonner, C. V., & Tay, L. (2023). Structural equation modeling in organizational research: The state of our science and some proposals for its future. *Annual Review of Organizational Psychology and Organizational Behavior, 10*(1), 495–517. https://doi.org/10.1146/annurev-orgpsych-041621-031401

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